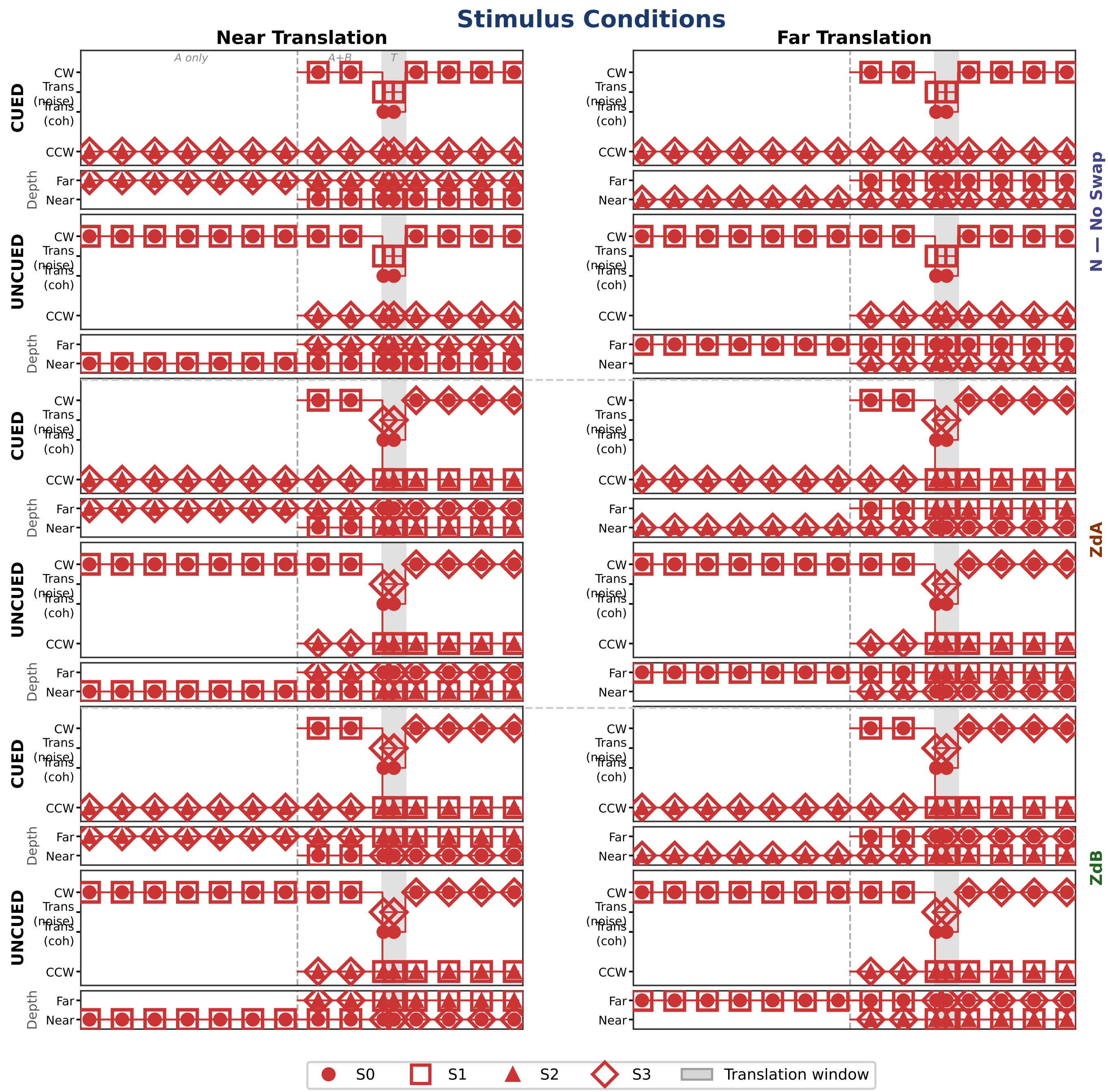
VRDots — Depth Swap Control (DepthSwapCtrl_005m)

Binocular: 260330_1853, 260331_0621, 260401_1313, 260401_1349, 260401_1541, 260401_1705 (n=1,152)   Monocular-R: 260330_2012, 260331_1530   Monocular-L: 260331_1705, 260331_1734 (n=769)

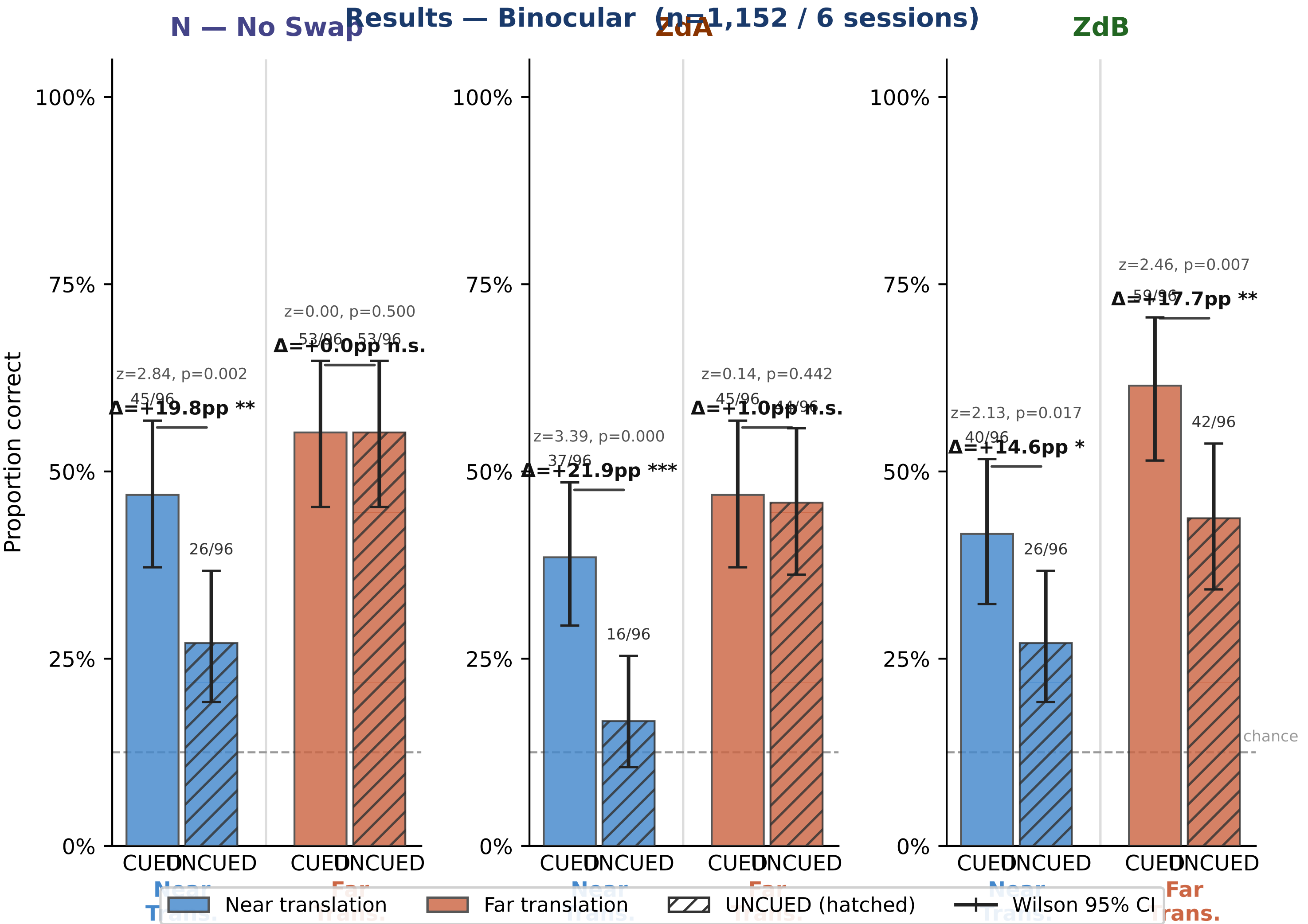
The DepthSwapCtrl experiment isolated the contribution of depth-plane identity to dot cueing by introducing mid-trial depth swaps at translation onset (T_s). Two swap types were tested alongside a no-swap baseline (N): ZdA swaps the depth planes of the two coherent dots (S0 and S2) at T_s, causing the cued coherent translator to change depth plane; ZdB swaps the two noise dots (S1 and S3) instead, leaving the cued coherent translator in its original depth plane. ZdA and ZdB are matched for the number of depth-plane transitions per trial, differing only in whether the transition affects the coherent translator. All sessions used a 0.05 m depth separation (~0.86° disparity at 2 m) and same-color stimuli (both fields red). Six binocular and four monocular sessions (2 right-eye, 2 left-eye) were collected from one observer.

Trajectory diagrams reflect the all-red stimulus (both fields red; same-color design). Columns are organized by depth of the translating field (Near | Far). S0●/S1□ = CW field (delayed in CUED); S2▲/S3◇ = CCW field. Filled markers = coherent subfields (S0, S2); open markers = noise subfields (S1, S3). Motion tracks are identical across swap types — the swap is depth-only at T_s. Depth tracks (thin row below each motion panel): in ZdA the coherent markers (S0●, S2▲) cross depth planes at T_s; in ZdB the noise markers (S1□, S3◇) cross instead.



VRDots — Depth Swap Control: Binocular Results

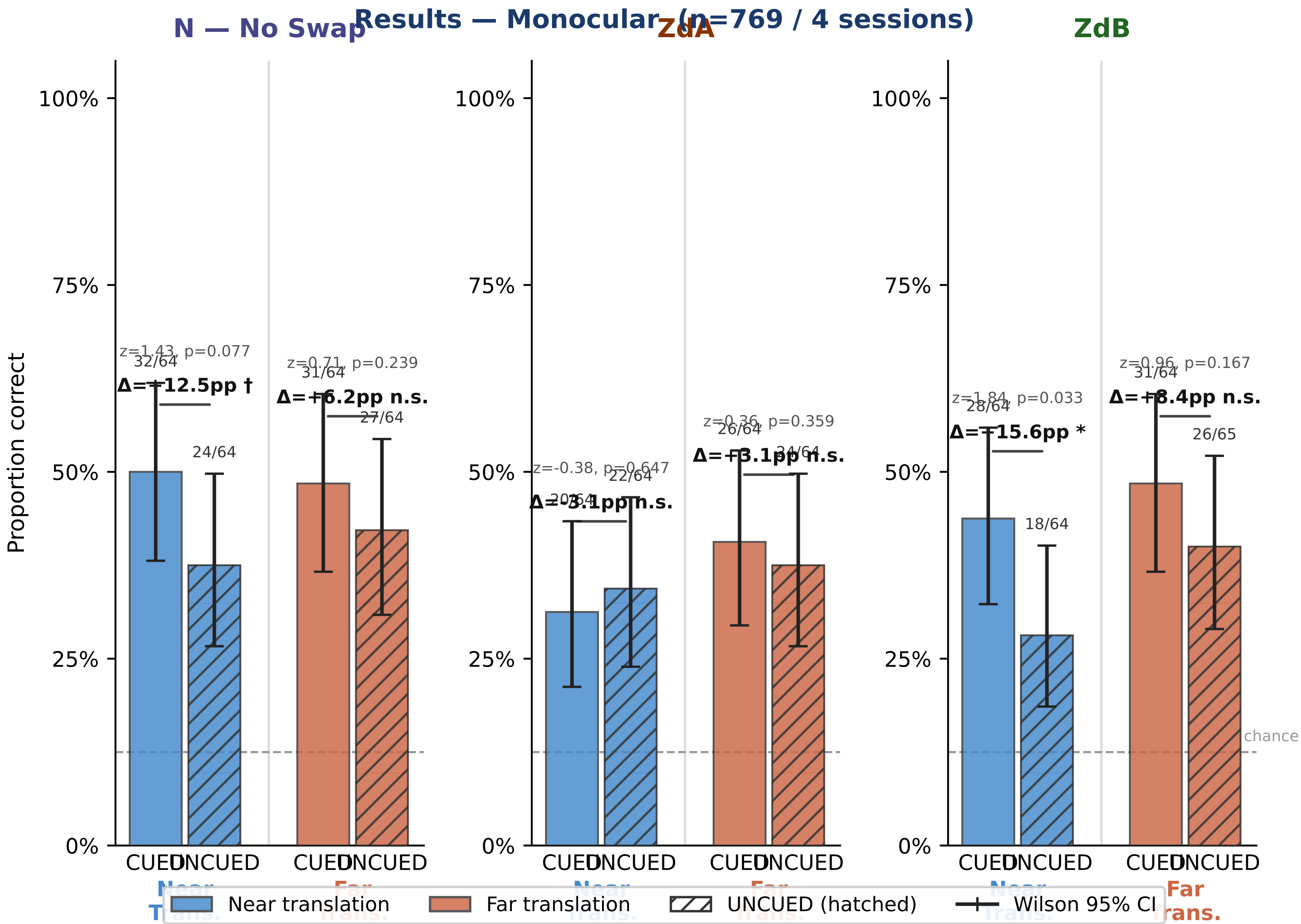
Analysis: Near/Far labels use the translating-field-depth convention. CUED Near = delayed field at Near (it translates Near). UNCUED Near = non-delayed field at Near (it translates Near; delayed field is at Far). Proportion correct pooled across color, rotation, and translation direction (≈32 trials/cell per swap×cond×depth within bino; ≈21 within mono). One-tailed z-test: CUED > UNCUED.



Findings

Binocular results (n=1,152, 6 sessions, 0.05 m depth separation): Dot cueing was reliable overall (+12.5 pp, z=4.30, p<.001). Using the translating-field-depth convention, Near translation yielded the larger cueing effect (CUED=42.4%, UNCUED=23.6%, Δ=+18.8 pp, z=4.79, p<.001) while Far translation showed a smaller marginal effect (CUED=54.5%, UNCUED=48.3%, Δ=+6.2 pp, z=1.50, p=.067). UNCUED performance is particularly low in the Near depth plane (23.6%), consistent with the anti-cued observer having difficulty detecting Near-plane translation. Far UNCUED performance was much higher (48.3%), leaving less room for cueing. Note: this translating-field-depth analysis reverses the apparent Near inversion that appears when using the delayed-field-depth convention (which conflates translating-field depth with cueing condition). Across swap types: ZdB produced the largest cueing (+16.1 pp, z=3.19, p=.001), followed by ZdA (+11.5 pp, z=2.33, p=.010) and N (+9.9 pp, z=1.95, p=.026). ZdA cueing is comparable to N binocularly, suggesting that depth-plane disruption of the cued translator alone does not produce large binocular cueing losses; ZdB's advantage reflects the benefit of keeping the cued translator in a stable depth plane while disrupting the distractor plane.

VRDots — Depth Swap Control: Monocular Results



Findings

Monocular results (n=769, 4 sessions): Overall cueing attenuated but survived (+7.1 pp, z=2.02, p=.022). Near translation cueing: +8.3 pp (z=1.69, p=.046); Far translation cueing: +5.9 pp (z=1.18, p=.12). The critical swap-type dissociation: ZdA collapsed to exactly +0.0 pp (z=0.00, p=.50) monocularly — the sharpest mechanistic result in the dataset. When binocular disparity is unavailable, changing the depth plane of the cued translator has zero behavioral consequence. ZdB survived monocularly (+12.0 pp, z=1.96, p=.025) and N was +9.4 pp (z=1.51, p=.066). ZdB's monocular survival suggests its advantage includes a monocular component, plausibly the brief positional shift of the non-coherent distractor at depth change (≈1.5–5 arcmin depending on eccentricity), which may disrupt the distractor surface representation independent of stereopsis.

Notes

Notes: (1) Sessions 260401_1541 and 260401_1705 were the 5th and 6th VR sessions in a single day (binocular) and show performance collapse (34–40% overall accuracy vs. 44–51% for other binocular sessions). Included in pooled statistics as planned. (2) Two color-related factors likely contribute to reduced cueing vs. two-color baselines (~28–50 pp): (a) absence of between-field color differences eliminates a reliable color-based field-identity cue present in all prior baselines; (b) the specific color used (red only) may independently reduce cueing if perceptual sensitivity to motion cueing is stronger for green than red for this observer — a single-observer asymmetry that cannot yet be ruled out. A red-only vs. green-only baseline session would dissociate (a) from (b). (3) The geometric confound in ZdA (0.05 m depth change induces 0–5 arcmin monocular positional shift at 2 m, scaling with eccentricity) cannot be fully dissociated from the stereoscopic depth-plane account at this sample size. The ZdA monocular null is consistent with either account.